Exceptional Architecture and Local Tourism Impact: A Synthetic Control Study of Hamburg's Elbphilharmonie

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Abstract

Exceptional architecture is widely recognized for its seeming impact on the cultural and economic trajectories of cities. However, the specific attractive potential of any given project and thus its impact on the regional economy is often contentious, and identifying measurable increases for outcomes of interest, such as visitor numbers, is empirically challenging. This study investigates the causal effects of the Elbphilharmonie concert hall in Hamburg on local tourism development. Employing the synthetic control method, we create a counterfactual scenario to estimate how Hamburg's tourism trajectory might have unfolded without the construction of the Elbphilharmonie. Contrasting recent debates highlighting the often underwhelming nature of 'star architecture', our findings indicate that the concert hall has had a major effect on tourism: In the eight and a half years from its opening in January 2017 to the second quarter of 2025, the Elbphilharmonie is estimated to be responsible for 18 million (7%) additional overnight stays. We conclude by discussing the impact of the covid pandemic and an identified sensitivity to the inclusion of Berlin as a comparison unit.

Keywords: exceptional architecture, tourism, synthetic control, Hamburg, Elbphilharmonie

1 Introduction

Exceptional architecture is known for its transformative power in shaping the economic and cultural landscape of its host city (Alaily-Mattar, Hall, et al., 2022; Alaily-Mattar, Johannes Dreher, et al., 2018; J. Dreher et al., 2023; Heidenreich and Plaza, 2015). In the context of heightened urban competition (E. Häusler and J. Häusler, 2024; Sklair, 2017; Zenker and Beckmann, 2013), especially among urban destinations (De Frantz, 2018), highly visible architectural projects can create recognizable regional identities and “act as a synecdoche for a city or region” (Alaily-Mattar, Hall, et al., 2022, p. 1), attracting tourists and inward investment in the process and thus providing a competitive edge (Balke et al., 2018; Scerri et al., 2019). Particularly the opening of the Guggenheim Museum in Bilbao in 1997 has brought the topic into the spotlight of scholarly attention: Its inauguration has been linked to amplified tourism, a revitalized local economy, and an elevated cultural reputation (Alaily-Mattar, Johannes Dreher, et al., 2018; Plaza, 2006; Plaza, Tironi, et al., 2009). However, the effectiveness of exceptional architecture in attracting tourists remains a subject of disagreement. According to Johannes Dreher, Alaily-Mattar, et al. (2020, p. 442), some studies emphasize the positive and lasting economic and tourism impacts, while others suggest that these effects are only short-lived and do not achieve the anticipated outcomes. We contribute empirical evidence to this debate in the form of a focused case study measuring the touristic impact of the ‘Elbphilharmonie’ in Hamburg. The Elbphilharmonie (Elbe Philharmonic Hall), inaugurated in January 2017, is a multifunctional building in Hamburg that includes concert halls, a hotel, apartments, and a publicly accessible plaza. Designed by the renowned architects Herzog & de Meuron, the building stands out by its wave-shaped glass façade atop a historic warehouse at Hamburg’s harbor (Johannes Dreher, Alaily-Mattar, et al., 2020; Hofmeister, 2021). It is recognized as one of Germany’s most notable recent architectural projects (Johannes Dreher and Thiel, 2022, p. 108) and is described as a “signature building” (Heuer and Runde, 2022, p. 90) or “tourist lighthouse” (Just, 2022, p. 55) that is “build to compete with New York, London or Sydney” (Zenker and Beckmann, 2013, p. 647). It also ranks as one of the costliest concert halls ever constructed worldwide (Heuer and Runde, 2022). Therefore, this study addresses the following research question: How did the inauguration of the Elbphilharmonie influence overnight stays in Hamburg’s local tourism sector?

To assess Elbphilharmonie’s causal effect on tourism in the city of Hamburg, we employ the synthetic control method (Abadie, 2021; Abadie and Gardeazabal, 2003) to measure the divergence of overnight stays against a counterfactual scenario representing tourism development in the absence of the Elbphilharmonie's construction. Traditional approaches like input-output methods or correlation analysis, as exemplified in the analysis of the economic and tourist impact of the Guggenheim Museum in Bilbao (Plaza, 2000; Plaza, 2006), are recognized as valuable for evaluating infrastructure benefits (Scerri et al., 2019). However, these methods have limitations in causally linking such benefits back to specific events (Doerr et al., 2020; Tkalec et al., 2017), often conflating different possible explanations, and in identifying effects unfolding over time (Johannes Dreher, Alaily-Mattar, et al., 2020). In this context, the synthetic control method serves as a suitable tool explicitly designed for measuring the impact of idiosyncratic events. Prior studies have successfully applied the synthetic control method to evaluate, e.g., the impact of European Union cohesion policies on tourism and culture (Brandano and Crociata, 2023), the influence of popular television series on (film) tourist arrivals (Tkalec et al., 2017), the effects of transport infrastructure on regional tourism (Albalate et al., 2023; Doerr et al., 2020), or the consequences of natural disasters on tourism and migration (Antonaglia et al., 2024; Kim and Lee, 2023).

Our empirical findings indicate that the Elbphilharmonie has had a sustained positive impact on tourism development in Hamburg, resulting in an additional 18 million overnight stays within eight and a half years since its opening. In the following sections, we will first outline the theoretical and empirical discussions regarding the influence of exceptional architecture on tourist attractiveness, while also addressing the methodological challenges associated with identifying causal effects. Section 3 presents the Elbphilharmonie as a case study, followed by a description of the data and the synthetic control method in Section 4. We then report the empirical results in Section 5, which includes the estimated 'Elbphilharmonie Effect' and its regional implications for tourism. Additionally, we will conduct robustness checks to validate our findings, concluding with a discussion of the broader implications.

2 Tourism and architecture

Over the past decades, tourism has evolved into one of the largest and fastest-growing industries globally, contributing substantially to the gross domestic product of many regions. For example, in 2019, tourism revenues in Europe amounted to a substantial 514.9 billion euros (World Tourism Organization, 2021). However, the pandemic triggered a sharp decline (Plzáková and Smeral, 2022; World Tourism Organization, 2023), as travel restrictions, testing requirements, and quarantines created lasting travel hesitancy among some individuals (Gössling and Schweiggart, 2022; McKercher and Tkaczynski, 2024). Nevertheless, and despite initial pessimistic forecasts, the tourism industry has undergone a robust recovery in most places, a phenomenon powered by and often referred to as "revenge tourism" (Meenakshi et al., 2024; Vogler, 2022). Once borders reopened, travelers were especially motivated to travel again, had in many cases accrued savings due to limited consumption, and had built long 'bucket lists' of target destinations (Meenakshi et al., 2024), with places renowned for their exceptional architecture being an appealing choice.

Tourism is also driven by increasing competition among urban centers for economic development (E. Häusler and J. Häusler, 2024; Sklair, 2017; Zenker and Beckmann, 2013), and many cities have invested in prestigious architectural projects to attract visitors (Andersson, 2014; Ponzini et al., 2020). When exceptional architecture stimulates new tourism, it directly and indirectly creates employment opportunities and additional tax revenues, which is in turn again economically beneficial for urban development (Alaily-Mattar and Thierstein, 2018; Ben-Dalia et al., 2013; Plaza, 2006). In this regard, exceptional architecture may act as a unique selling point that adds a layer of urban prestige (Andersson, 2014; Romão, 2025) and often becomes a representative symbol of urban progress (Balke et al., 2018; Boniface et al., 2021; Scerri et al., 2019; Specht, 2014). This aligns with the findings from Plaza, Esteban, et al. (2024), who link the effects of exceptional architecture with findings from cognitive psychology and social cognition (DiMaggio, 1997). They emphasize that such architecture holds major cultural and symbolic meanings, as they can create strong symbolic associations and are easily remembered due to their distinctiveness, elevating the global visibility of their locations and making them memorable to potential visitors and media consumers. These architectural icons, therefore, can serve as crucial connectors in the global media landscape, reinforcing the prominence and boosting the attractiveness (for visitors) of their host locations. Taking a leading role and gaining visibility is especially important in the context of post-pandemic tourism, as the recovery intensifies competition among tourism-dependent economies to attract visitors (Abbas et al., 2024). While much of the literature focuses on newly built flagship projects, recent examples of exceptional architecture (e.g., Zeitz Museum of Contemporary Art Africa, Cape Town or Fondazione Prada, Milan), including the Elbphilharmonie, combine new design with the adaptive reuse of existing structures. Such hybrid projects integrate historical continuity with contemporary architectural expression, reinforcing urban identity and tourist appeal through both renewal and preservation (Alaily-Mattar and Thierstein, 2018; Ponzini et al., 2020). The above deliberations support the expectation that the Elbphilharmonie had a positive effect on tourism in Hamburg (as measured by overnight stays), the estimation of which is the primary goal of this study.

While there are many plausible ways in which exceptional architecture can spur tourism growth and thus regional development, measuring its precise impact remains challenging and studies of other flagship projects have yielded mixed results. For example, studies of the 'Guggenheim Effect' show that museum visitors played a major role in the growth of tourism in the Basque Country after the opening (Plaza, 2000). However, there are methodological hurdles in determining the actual effects. In the specific case of the Guggenheim Museum, idiosyncratic factors such as the dissolution of the terrorist organization ETA, general tourism developments, economic cycles, the increasing attractiveness of neighboring regions, and comprehensive urban developments must be taken into account, making it challenging to isolate the phenomenon (Plaza, 2000; Voltes-Dorta et al., 2016). Another example underscores the difficulty of capturing the causal effects of exceptional architecture on tourism. The study by Johannes Dreher, Alaily-Mattar, et al. (2020) examines the tourist effect of the Kunsthaus Graz, Austria (an architecturally exceptional art exhibition center inaugurated in 2003) based on the number of museum visitors on the number of overnight stays in the city. The results show that, despite a decline in visitor numbers to the museum, overnight stays in the city increased. The authors characterize the study's outcome as a "rather sobering finding" (p. 449), since they discovered no statistical proof indicating that Kunsthaus Graz played a direct role in the upswing of overnight tourism. However, they posit that the impact on tourism is more attributed to the architecture itself rather than the exhibitions because the museum operates as an icon representing the city, drawing in tourists without necessarily motivating them to explore the art exhibitions. Our study builds on this assumption by using the Elbphilharmonie in Hamburg as a case study to explore the forward linkage effects (Gelman and Imbens, 2013; Hiller, 1998) of its inauguration to address the research question of how exceptional architecture can influence overnight stays in the local tourism sector.

3 The Elbphilharmonie

Germany, a major player in the global tourism export market (Gössling and Schweiggart, 2022), has faced severe impacts from the COVID-19 pandemic, particularly in its hospitality sector. In this context the case of Hamburg becomes particularly interesting as it features the architectural masterpiece, the Elbphilharmonie. As a relatively recent addition to the city's skyline, it may have emerged as a popular destination for travelers seeking unique experiences, especially following the easing of pandemic restrictions. From the outset, the Elbphilharmonie was ascribed identity-forming characteristics and recognized for its potential socio-economic influence due to its location, as well as the exceptional visual and emotional appeal (Alaily-Mattar, Akhavan, et al., 2021). Architecturally integrated in/on the brick corpus of a former cocoa warehouse in Hamburg's harbor area, it is adjacent to the historic warehouse complex and UNESCO World Heritage Site 'Speicherstadt and Kontorhaus District with Chilehaus' (in the following: Speicherstadt) (Heuer and Runde, 2022) (see Figure 1). Situated at HafenCity's western edge, the Elbphilharmonie is within walking distance of major attractions such as the Speicherstadt, the Landungsbrücken waterfront, and Hamburg's theater musical dome across the river, forming part of a geographical concentration of cultural sites typical of urban tourism areas (Ashworth and Page, 2011).

The constructions were completed in November 2016 and officially opened in early January 2017. The striking design features a conspicuous iceberg/wave structure made of glass that stands 110 meters tall. Designed by internationally renowned architects Herzog & de Meuron, who also created the Tate Modern in London and the 'Bird's Nest' Olympic Stadium in Beijing, the building is situated in a prominent location (Alaily-Mattar, Bartmanski, et al., 2018), and—on par with comparable iconic architectural landmarks (Ghalejough et al., 2024)—has sparked extensive media coverage. These factors qualify it as a distinct type of exceptional architecture known as 'star architecture' (Johannes Dreher and Thiel, 2022). But the Elbphilharmonie also shares another attribute often associated with star architecture: it is "expensive and controversial" (Alaily-Mattar, Hall, et al., 2022, p. 3). Initially estimated at 187 million euros, the construction cost eventually rose to 866 million euros, making it one of the most expensive concert halls ever built (Hofmeister, 2021). During the selection process, there were some controversies, notably the awarding of the design contract to architects Herzog & de Meuron without an open bidding process or architectural competition (Alaily-Mattar, Akhavan, et al., 2021, p. 101).

From a functional perspective, the structure serves as a concert hall, which also includes a hotel and private residences. Moreover, situated at a height of 37 meters on top of the brick corpus, it encompasses an open plaza accessible to the public, offering residents and visitors an expansive view of the city (Holz, Alatur, et al., 2023). By early 2024, this plaza has been experienced by over 20 million visitors (DPA, 2024), while within a single year until mid-2024, 852,000 tickets were sold for the concert hall (Holz, Schulz, et al., 2024). Nevertheless, the Elbphilharmonie project is linked to a 'Hamburg Effect' instead of a 'Guggenheim Effect' (Heuer and

Runde, 2022). Unlike the Guggenheim Museum in Bilbao, which was constructed as a catalyst for the regeneration of a "rundown" (Heidenreich and Plaza, 2015, p. 1441) industrial city and subsequently evolved into a major tourist attraction with tangible economic benefits and a thriving art and creative scene (Plaza, Tironi, et al., 2009), the Elbphilharmonie was designed with the primary goal of reinforcing Hamburg's identity as a cultural and musical hub (Heuer and Runde, 2022). The focus to serve the local and national population gets evident in the origin of the audience attending performances in the concert hall, where the proportion of foreign attendees has not surpassed 6%. In contrast, approximately 60 to 70% of visitors to the Guggenheim Museum in Bilbao come from foreign countries (Heuer and Runde, 2022, p. 97). While it is impossible to eliminate all contextual influences, the lack of major new redevelopment projects coinciding with the Elbphilharmonie's opening strengthens confidence that the tourism increase is mainly linked to the building itself. Its opening in 2017 took place within HafenCity, a long-term regeneration project started in the late 1990s and developed gradually over decades, rather than as a single transformative event (Landis, 2022). Hamburg, despite a slow decline in port activity, remained one of Europe's wealthiest cities with a growing population, and the Elbphilharmonie was designed not as an economic recovery tool but as a cultural landmark (Heuer and Runde, 2022, p. 96). In urban economics literature, such gradual processes are unlikely to obscure causal inference, as their impacts are absorbed over time rather than centered at a single point (Baum-Snow and Ferreira, 2015; Sun et al., 2023), and are therefore differenced out by the synthetic control through pre-treatment trajectory matching (see Section 4).

To study the impact of exceptional architecture on local tourism, using the Elbphilharmonie in Hamburg as an example, is beneficial in two ways. Firstly, the relatively low number of foreign concert attendees suggests that, if there was a positive effect, it is the architectural attraction itself rather than the concerts that attracts tourists, providing empirical support for the assumptions of Johannes Dreher, Alaily-Mattar, et al. (2020). Secondly, when measuring the touristic development in Bilbao in the 1990s, multiple factors beyond the opening of the Guggenheim Museum contributed to the city's attractive for tourists, including extensive urban redevelopment of former industrial areas, economic diversification, and a decline in ETA-related violence (Keating and De Frantz, 2003; Gómez and González, 2001). Or, as Hassink et al. (2025) frame it, "the megaproject of the museum and its surrounding area [. . .] became the symbol of urban change but, arguably, it was not its cause" (p. 257). In contrast, during the opening of the Elbphilharmonie, Hamburg did not experience similar large scale idiosyncratic contextual factors (for a discussion, see

Section 5), making it more feasible to accurately estimate the isolated causal effects of exceptional architecture on tourism.

Figure 1: (a) Elbphilharmonie Hamburg with its wave-shaped glass façade © T. Rätzke (b) View over the Speicherstadt © T. Rätzke (c) Architecture and construction details © Herzog & de Meuron (d) Brick corpus of the cocoa warehouse in 1967 © Zoch. Source: Elbphilharmonie & Laeiszhalle (2024) (e) Location of the Elbphilharmonie and nearby attractions. Modified base map from Freie und Hansestadt Hamburg LGV (2025).

4 Data and Method

4.1 The synthetic control method

In this paper, we use the synthetic control method (Abadie, 2021) to assess the touristic impact of the Elbphilharmonie. The synthetic control method aims to answer the question of whether a discrete intervention (here, the building of the Elbphilharmonie) had a causal effect on some quantity of interest (here, overnight stays), i.e., whether there was an increase (or decrease) in the quantity of interest which would not have occurred without the intervention. In pursuit of this goal, the method follows a comparative case study design which contrasts the unit for which the intervention occurred (Hamburg) with a set of comparable units where the intervention did not occur (other cities). More specifically, the synthetic control method proceeds by constructing a weighted average of contrast cases representing a 'synthetic' version of the intervention unit, which can be used to assess counterfactual scenarios. Weights are determined in a way as to maximize the pre-intervention similarity of the synthetic and the real unit. Similarity is determined based on a set of predictors, which themselves have associated importance weights also subject to optimization. Weight optimization follows the nested convex optimization procedure outlined in Abadie, Diamond, et al. (2010) and implemented in the `synth` / `tidysynth` R packages (Dunford, 2025). This procedure estimates weights for the control units as well as the predictors, where unit weights minimize the distance between the predictors of the intervention unit and the predictors of the control units and predictor weights maximize the pre-intervention outcome fit. Given sufficient pre-intervention similarity (taken to indicate that the behaviour of the quantity of interest is well-approximated by the weighted average of the control units), post-intervention deviations between the synthetic and the real case are then interpreted as a consequence of the intervention.

An important assumption is the absence of other idiosyncratic shocks that could lead to a deviation of the treated unit in terms of the outcome of interest. While there was a considerable shock to tourism throughout the observation period in the form of the COVID-19 pandemic, this was not unique to Hamburg but more or less universal. Nevertheless, the pandemic led to considerable variability, which can make the identification of a clear signal more difficult. Accordingly, it is important to check the robustness of the causal interpretation of a treated-control divergence, which can be achieved through two different kinds of placebo tests: First, placebos can be constructed by assigning the intervention to other units than the one for which it occurred. Second, instead of switching the unit, the analysis can be rerun with the intervention shifted to a different point in time. Here, we conduct such a temporal placebo trial by setting the intervention date to the year 2010 instead of 2017. If these placebo trials yield similarly large post-intervention differences, results are likely to represent noise and should be interpreted with care. If these simulations yield similarly large effects as the actually treated unit, results are again deemed not robust. In addition to these placebo test, we conduct a leave-one-out test (Abadie, Diamond, et al., 2010) by iteratively removing cities from the donor pool and rerunning the weight optimization procedure. Deviations in the resulting match signal sensitivity to individual members of the donor pool, which need to be evaluated qualitatively. All analyses were conducted with the `tidysynth` package for the `R` programming language (Dunford, 2025).

4.2 Predictors and control cases

Our primary variable of interest is the number of overnight stays, observed at quarterly level from 1998 to Q2 2025. Compared to other measures of touristic activity, such as arrivals, the number of overnight stays was the most complete series, and preliminary tests indicated that arrivals did not yield substantively different findings. The latter is not surprising, as the average length of stay (overnight stays divided by arrivals) for city breaks has remained comparatively constant over time. As primary predictors for the identification of weights that minimize pre-intervention differences between the observed series of overnight stays and the weighted average of the control units constituting the synthetic series, we then use averages for all 2-year pre-intervention periods, separated by quarter (i.e., different averages for each quarter for each period). This specification allows for different periodicity patterns (e.g., coastal vs. alpine tourism) to influence the selection of optimal weights. In addition, we also use 5-year periodic averages for per capita GDP as well as population size in the three years preceding the intervention as predictors. GDP per capita and population size are valuable additional predictors for constructing the synthetic Hamburg as they capture the key economic and demographic fundamentals that influence tourism demand and ensure the control units reflect similar structural conditions prior to the intervention.

For the candidate pool of control cases we selected 18 German cities, including Berlin, Munich, and Cologne as comparatively sized candidates, as well as Rotterdam, Amsterdam, Copenhagen, and Helsinki. The German units capture national touristic trends and mostly uniform policy responses during the pandemic, while the international peers account for tourism dynamics specific to Northern European maritime hubs. Rotterdam, in particular, serves as a primary structural analog due to its shared identity as a major harbor city transitioning toward a culture-driven waterfront economy. Together, the international units represent recognized functional and economic benchmarks for Hamburg (Alberti et al., 2023). Within the city break market, these destinations offer a comparable urban image that appeals to the creative and cosmopolitan consuming classes (Maitland, 2010), and so function as potential spatial substitutes for visitors (Çakar et al., 2025). This focused pool minimizes cross-country confounding variables, such as diverging travel restrictions, which would otherwise introduce noise into the synthetic series during the COVID-19 period. Inclusion of further cities or removal of the included smaller German cities did not lead to significant differences in the findings, supporting the selection reported here.

4.3 Data sources

The data for German cities were obtained from two primary sources. Initial data were drawn from the online portals of the respective federal state statistical offices, where certain information is available in digital formats. Because the data available from the public interfaces were incomplete, we supplemented this initial effort with archival data obtained through a cross-regional request to the federal statistical offices. This request was managed by the Bavarian State Office for Statistics, which coordinated with the other statistics offices to consolidate a comprehensive dataset. To ensure structural comparability, the donor pool was limited to urban centers. Smaller cities were excluded because their scale and functional composition differ substantially, making them unsuitable to approximate a metropolitan tourism economy (Abadie, 2021; Abadie, Diamond, et al., 2015). Furthermore, we had to exclude most East German cities because, in exchange with the state statistical office, we were informed that these cities underwent substantial boundary changes and shifts in publication practices over time, which prevents the construction of consistent long-term data series.

For Munich and Nuremberg, pre-2005 data was only available on a yearly basis. For these cases, we interpolated quarterly data by using the post-2005 share for each quarter (which was checked to be sufficiently stable over time) and then distributing the available yearly data accordingly. Data for cities outside Germany were collected directly from the online platforms of the respective national statistical offices (Statistics Netherlands, Statistics Denmark, and Statistics Finland). Data on GDP and population were obtained from Eurostat for the corresponding NUTS3 region.

5 Results

5.1 Tourism development in Hamburg

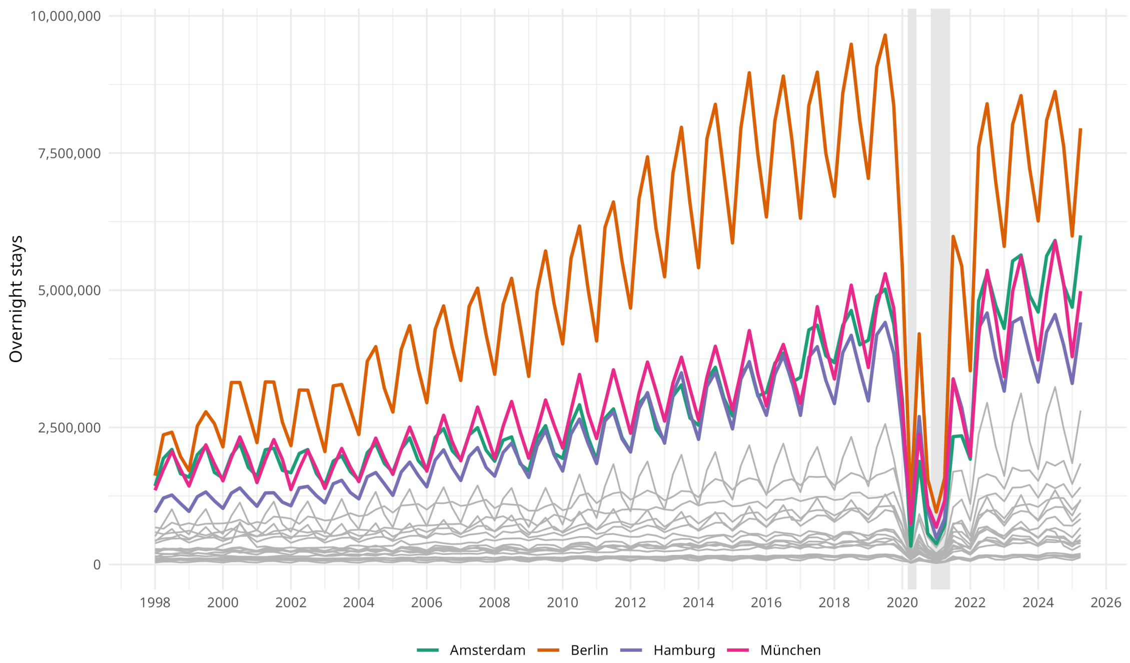

Figure 2: Quarterly number of overnight stays for Hamburg and the 22 sampled control cases. Hamburg and selected comparable cities are highlighted. The grey shaded areas mark major pandemic-related lockdowns from March to May 2020 and from late 2020 to May 2021.

Hamburg has experienced a steady increase in tourism over the past 30 years, mirroring trends observed in other major German and Northern European cities. Figure 2 shows the quarterly number of overnight stays for Hamburg and the 22 other sampled cities, revealing a growth trajectory that aligns closely with cities such as Munich or Amsterdam. From 1998 to 2025, overnight stays in Hamburg more than tripled, rising from approximately 4.5 million to nearly 16 million. Among the highlighted cities, Berlin stands out with the steepest increase, particularly from the early to mid-2000s. This growth can be attributed to Berlin's evolving role as the capital of reunified Germany, its emergence as a global cultural and political hub, and a surge in international conferences, exhibitions, and events. Munich also exhibits strong growth, supported by its prominence in the business travel sector and large-scale cultural events. Amsterdam shows consistent increases in demand, though this has been more recently tempered by regulatory constraints. The city's 'Tourism in Balance' policy, introduced in 2021, caps annual tourist hotel overnight stays at 20 million in response to concerns over overtourism and resident wellbeing

(City of Amsterdam, 2025).

In contrast, Hamburg has traditionally lacked the same scale of recurring mega-events or administrative centrality. While it today has a strong tourism base and a vibrant cultural sector, its international profile has historically been more modest. A notable moment in the data is the dip around 2001 across several cities, particularly Berlin, Amsterdam, and Munich. This downturn likely reflects the global travel shock triggered by the 9/11 terrorist attacks, which disproportionately affected internationally exposed destinations. Hamburg, whose tourism at the time was more domestically oriented, appears less impacted by the immediate fallout. Despite Hamburg's steady long-term growth, these descriptive trends alone are insufficient to isolate the effect of a single intervention such as the construction of the Elbphilharmonie. To more precisely evaluate its impact, we turn to the synthetic control method in the following section.

5.2 The Elbphilharmonie effect

Table 1: Pre-intervention comparison of selected predictors

Variable	Hamburg	Synthetic Hamburg	Donor pool mean
Overn. stays 2010–2011 Q1	1,771,832	1,845,242	724,781
Overn. stays 2010–2011 Q2	2,498,832	2,428,625	940,217
Overn. stays 2010–2011 Q3	2,716,609	2,648,949	1,057,967
Overn. stays 2010–2011 Q4	2,250,692	2,329,089	913,615
Population 2009–2010	1,773,162	1,586,950	714,265
GDP 2005–2009	53,460	51,122	44,821

Note: All values are averages over the indicated period. Synthetic Hamburg is the weighted average of control cases using the optimized weights (where Berlin, Düsseldorf, Cologne, Stuttgart, and Hannover were selected as relevant donors), while the last column reports the simple average of all donors.

The synthetic control method achieves a reasonably close pre-intervention fit between the synthetic unit and the real Hamburg, with a root mean square predictive error (RMSPE) of 70354.8 (compared to observed quarterly overnight stays in the millions). Based on these weights, it finds a positive divergence after the intervention, with an estimated surplus of 18.1 million overnight stays in the 8.5 years between the opening of the Elbphilharmonie in January 2017 and the end of the observation period in Q2 2025, compared to the counterfactual Hamburg without Elbphilharmonie. Figure 3 shows the quarterly series of overnight stays for Hamburg and its 'synthetic twin' (a) and the difference between the two (b). Despite the volatility due to the COVID-19 pandemic, there is a clearly visible divergence in the period following the

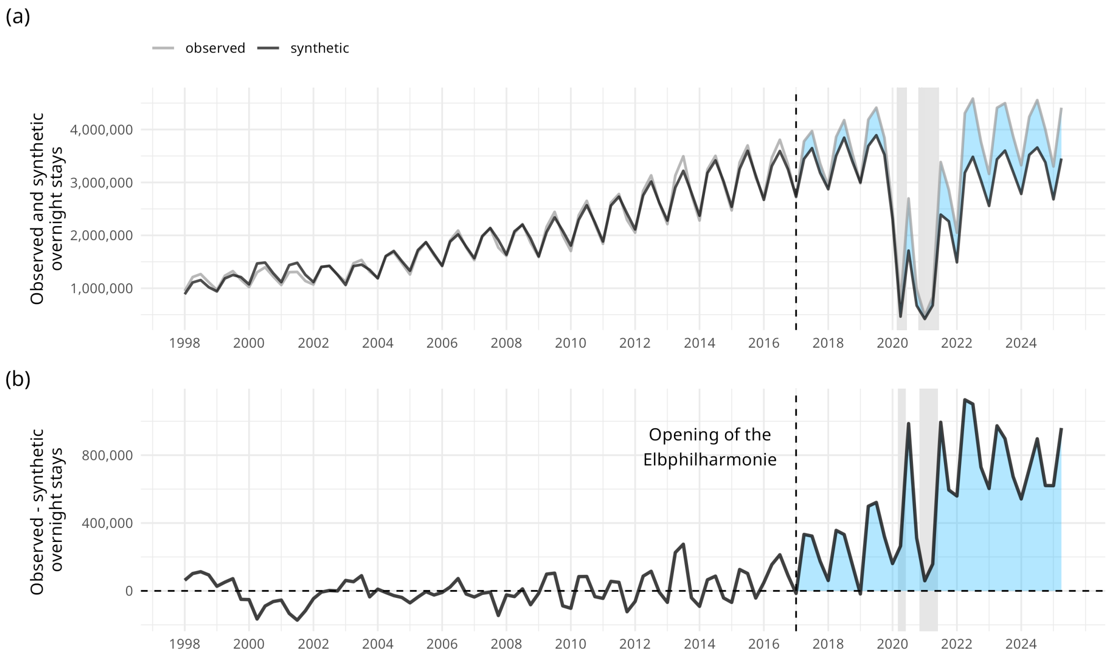

Figure 3: (a) Synthetic and observed series of overnight stays, (b) difference between synthetic and observed series. Blue shaded areas highlight the post-treatment difference between the synthetic and observed series. The grey shaded areas mark major pandemic-related lockdowns from March to May 2020 and from late 2020 to May 2021.

intervention, with a pronounced increase in overnight stays especially in the period after the end of the pandemic. The difference peaks in Q2 2022, where an extra 1,126,533 overnight stays are recorded. Vice-versa, troughs of the difference plot align with pandemic-induced lockdowns in Germany (as indicated by the shaded areas in Figure 3) and are followed by peaks. This might be an indication of the Elbphilharmonie—still being relatively recent at the time of the pandemic—contributing to Hamburg attracting 'revenge tourism', i.e., tourism driven by a disproportionate willingness to travel after the lockdown (Vogler, 2022). Whether recent landmarks and the visibility they create systematically interact with such rebound effects is however speculative at best and beyond the scope of this study.

The synthetic control method is only meaningful if a sufficiently good pre-intervention match between the synthetic and the observed outcome series can be achieved. In addition to the previously mentioned RMSPE as well as the visual guidance to this effect provided in Figure 3, Table 1 provides numerical comparisons between the observed and synthetic units and the donor pool average for overnight stays (2007 to 2011 period), as well as GDP and population statistics.

While the donor pool average is far below Hamburg in both the outcome variable (overnight stays) as well as the additional predictors (per-capita GDP and population), the weighted average of the synthetic unit comes quite close, with an error of less than 5% for most predictors. Regarding the composition of the estimate, the optimization procedure allocates most of the weight to five cities: Berlin (0.31), Düsseldorf (0.29), Cologne (0.18), Stuttgart (0.17), and Hannover (0.06). Although the selection of large German cities as primary donors seems generally sensible, interpretability of specific weights is known to be a limitation of the synthetic control method (Abadie, 2021). Similarly, variable importance weights are available for inspection but somewhat difficult to assess: The most important predictors are the numbers of overnight stays at the start of the observation period, i.e. at the end of the 1990s and in the early 2000s. Overall, Q4 is featured somewhat more often at the top of the list than the other quarters for identifying a good match for Hamburg's pre-intervention tourism dynamics, possibly representing specific seasonal dynamics as an important matching criterion. GDP and population size are less important, with population in the 2005 to 2009 period as the highest-weight non-overnight-stays predictor in position six.

A more accessible metric is the ratio of post-intervention to pre-intervention mean squared predictive error (MSPE, Table 2), which indicates how much better the synthetic series matches the observed series before vs. after the intervention. Given a significant effect of the intervention, this should be large for the treated unit but not for the control units. This is indeed the case for Hamburg, with a post-intervention error that is more than 50 times higher than before the intervention, clearly indicating a change-point in the tourism dynamics around the opening of the Elbphilharmonie.

5.3 Robustness checks and placebo trials

While the synthetic control research design is not amenable to classical statistical hypothesis tests, the robustness of the results can nevertheless be assessed by conducting 'placebo trials'. This is arguably especially important in a case like the one discussed here, where a global shock in the form of the covid pandemic introduces uncertainty into the studied system. We accordingly conduct two kinds of placebo trials: First, a unit-switching placebo trial which implies running the synthetic control method for units that did not receive the intervention (Abadie, Diamond, et al., 2015). And second, a backdating placebo trial, pretending that the intervention took place at an earlier point in time than the actual intervention (Abadie, 2021).

If effects similar in size to the one identified by running the synthetic control method for the correct unit at the correct time can be obtained by placebo trials,

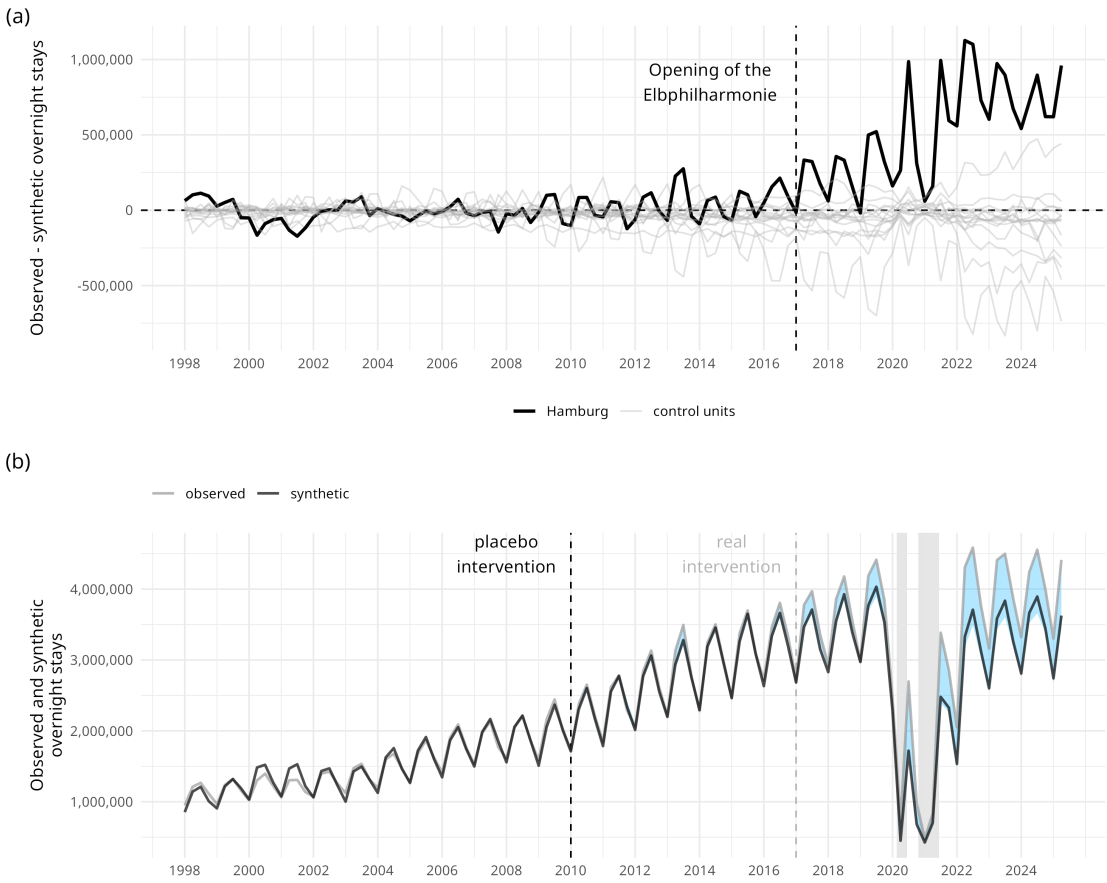

Figure 4: (a) Difference between observed and synthetic overnight stays for Hamburg (black line) and placebo test with control units (grey lines). Only control units within 2 times the MSPE of Hamburg are shown to avoid visual clutter (see Abadie, Diamond, et al. 2010). (b) Temporal placebo test showing the synthetic and observed series with the intervention predated to 2010.

the result is deemed not robust. Figure 4 (a) shows the results of the unit-switching placebo trial with the analysis repeated for all control units in the donor pool. While there is a significant amount of variation for the control unit trials, none of them match the same effect magnitude as the one identified for Hamburg. The closest control is Rotterdam, which is the only control unit with a considerably elevated ratio of post-intervention to pre-intervention MSPE (Table 2). This can be attributed to the fact that Rotterdam underwent its own 'intervention' during the same period, marked by a boost of its touristic appeal through progressive architecture (e.g. the Markthal or Depot Boijmans Van Beuningen) and cultural events (Nientied, 2021). However, the difference series for Rotterdam shows considerable fluctuation even after the end of the covid pandemic and its characteristics should probably not be overstated.

The backdating robustness check also gives no indication that a similar effect

Table 2: Sensitivity statistics based on placebo and LOO tests

	Pre/post intervention	Leave-one-out test (LOO)	
City	RMSPE ratio	Pre-intervention RMSPE	Treatment Effect
Hamburg	51.03	70,355	18,127,946
Rotterdam	43.56	70,355	18,128,204
Bremen	32.3	70,355	18,127,746
Helsinki	10.97	70,355	18,127,769
Bochum	9.98	70,355	18,129,056
Köln	9.07	67,157	15,040,720
Nürnberg	8.78	70,355	18,127,842
Dortmund	8.53	70,354	18,129,988
Hannover	6.09	67,228	16,430,435
Kopenhagen	5.29	73,892	16,198,899
Münster	4.37	70,354	18,128,126
Amsterdam	4.29	70,355	18,128,434
München	3.79	70,355	18,129,079
Bielefeld	3.09	70,355	18,128,320
Bonn	2.51	70,355	18,128,235
Duisburg	1.85	70,355	18,129,146
Berlin	1.56	249,660	3,214,405
Wuppertal	1.47	70,355	18,127,804
Bremerhaven	1.27	70,355	18,129,305
Düsseldorf	1.01	68,505	15,283,764
Stuttgart	0.95	70,725	18,638,253
Essen	0.93	70,355	18,128,364

Note: Pre-intervention RMSPE and treatment effect for Hamburg refer to the baseline estimate. Berlin is highlighted because the estimate is sensitive to its exclusion from the donor pool.

to the one obtained for the actual intervention can be recovered from a temporal placebo. Figure 4 (b) shows the result of running the synthetic control procedure with the intervention advanced from 2017 to 2010: The synthetic series does not significantly diverge from the observed series between the placebo intervention and the actual intervention, but does still recover the divergence observed after the opening of the Elbphilharmonie in 2017. Together, these results make us confident that the observed divergence is not purely a result of noise.

Next to the placebo trials presented above, we also run a leave-one-out sensitivity test where we iteratively remove each city contained in the donor pool and recompute the optimal weights on the remaining $n-1$ donors (Abadie, Diamond, et al., 2015). For every fit generated this way, Table 2 contains the pre-intervention RMSPE as a measure of fit and the treatment effect (i.e., the total post-intervention difference between the synthetic and the observed unit). This leave-one-out test reveals a sensitivity of the result to the inclusion of Berlin – without Berlin, a pre-intervention match as close as the one including Berlin cannot be achieved. One factor of this is likely technical, since Berlin is the only city in the donor pool that is larger than Hamburg and one of few that matches Hamburg's overnight stay count. Since the synthetic unit has to be within the convex hull of donors, the options are limited without Berlin in the pool. Instead, the estimate excluding Berlin as a potential donor resorts to Munich as a primary large-city donor, which is a bad match for Hamburg's pre-intervention trajectory of overnight stays. It is also substantively plausible that tourism growth in Hamburg is better represented by Berlin rather than Munich, as the latter is culturally more distinct and has an idiosyncratic touristic profile due to the Oktoberfest.

A final potential source of confounding is the designation of the Speicherstadt, located in direct proximity to the Elbphilharmonie, as a World Heritage Site in 2015. This carries the risk of misattributing a potential World Heritage related effect on overnight stays to the Elbphilharmonie. To assess the extent to which the World Heritage nomination might have generated an attention pattern similar to that of the Elbphilharmonie, we rely on Google Trends data (Google Trends, 2025). Figure 5 presents the search indices for the terms "Elbphilharmonie" and "Speicherstadt" and shows a clear difference in visibility: while there is a notable spike around the Elbphilharmonie's opening, search behaviour reacts only minimally to the World Heritage nomination. Although Google Trends measures search interest rather than actual visitation, prior research has shown that it is a reasonable proxy for destination demand (Dinis et al., 2019; Önder, 2017; Botha and Saayman, 2024). The observed difference supports the interpretation that the Elbphilharmonie, rather than the World Heritage designation, primarily drove the increase in interest relevant to Hamburg's tourism.

6 Discussion

This study uses a synthetic control approach to estimate that the opening of Hamburg's Elbphilharmonie in 2017 generated approximately 18 million additional overnight stays over the following 8.5 years. To contextualize these findings, an illustrative projection based on the average on-site travel expenses of €129 per person per day in 2023 (Hamburg Tourismus, 2024) suggests approximately €2.3 billion in additional touristic spending by overnight visitors. Although this simple calculation is not a formal economic model, it provides a rough indication that, despite the at-the-time heavily criticized project cost of around €870 million (Maak, 2015), the expense could have been broadly offset at the macroeconomic level within a relatively short period. Although our analysis is in line with the Elbphilharmonie having been an economic success for the city of Hamburg, it doesn't indicate that this need be the case for all architectural lighthouse projects. Indeed, many case studies outline how such projects failed to meet economic or cultural expectations, such as the PalaFuksas in Turino (Vanolo, 2022), the Phæno in Wolfsburg (Alaily-Mattar, Büren, et al., 2019), or the Oscar Niemeyer Centre in Avilés (Somoza Medina, 2019).

As discussed earlier, the observation period overlaps the COVID-19 pandemic, which had severe consequences for tourism worldwide and thus might plausibly interfere with the estimation of the Elbphilharmonie effect on tourism. However, we believe the effect visible in this study to be more than just a pandemic-induced artefact for two reasons: First, the effect is already visible for the post-treatment period before the onset of the pandemic in 2020. Second, the donor pool is almost entirely composed of German cities, and only German cities received significant weight, which limits the influence of cross-country confounding from travel and policy differences especially during the COVID-19 crisis. Although some variation across federal states existed, Germany followed a broadly coordinated pandemic policy timeline (e.g., the March 2020 contact ban, the mid-December 2020 winter tightening, and the April 2021 federal emergency brake) (Bönisch et al., 2020; Schuppert et al., 2021). This reduces the likelihood of bias due to non-parallel recovery trends. Nevertheless, the possibility that pandemic induced revenge-tourism led to an amplification of the effect of lighthouse projects such as the Elbphilharmonie remains a challenging limitation of the approach presented here.

Another limitation of this study is the sensitivity of the result regarding the inclusion of Berlin in the donor pool. The inclusion of Berlin is required to achieve a satisfactory pre-intervention fit between the synthetic and observed Hamburg—its removal deteriorates both the pre-intervention RMSPE as well as the treatment effect, as shown by a leave-one-out test. While this does not present evidence against the Elbphilharmonie impact (as would a finding of no post-intervention difference with a good pre-intervention fit), it limits the generalizability of the difference between the observed and synthetic Hamburg to an arbitrary pool of comparison units. On the other hand, it is maybe not all that surprising that Berlin, the closest comparably sized city, is an important point of comparison. Its importance in reconstructing the trajectory of Hamburg's touristic growth is in our view thus not an irrefutable signal of invalidity but nevertheless important for contextualizing the primary findings of the study.

Finally, exceptional architectural projects can evoke complex social effects on adjacent neighborhoods beyond their direct economic impact. These may include changes in the local housing market, shifts in resident demographics, and gentrification, as noted in earlier research (Balke et al., 2018; Novy and Colomb, 2019). Additionally, an increased volume of tourism can lead to overtourism that exceeds local social capacities and disrupts residents' everyday social life (Namberger et al., 2019).

7 Conclusion

This study examines the local impact of exceptional architecture on tourism by analyzing the inauguration of the Elbphilharmonie in Hamburg. Using the synthetic control method, we construct a data-driven counterfactual scenario to assess whether the construction of the Elbphilharmonie led to an increase in overnight stays that would not have occurred otherwise. Our analysis provides evidence that the Elbphilharmonie generated an estimated 18 million additional overnight stays in the eight and a half years following its opening in 2017. This study contributes to the literature on the effects of exceptional architecture by providing causal evidence of a lasting tourism impact, supporting earlier qualitative findings and offering empirical clarity to a field often characterized by mixed results. The robustness of these findings, even in the face of pandemic-induced shocks, underscores the potential for flagship cultural projects to drive urban tourism.

However, while this study reports a major tourism impact, the findings remain subject to limitations regarding donor pool sensitivity and the potential for pandemic-induced trends to influence the post-2020 estimates. As detailed in the discussion, these factors necessitate careful contextualization of the counterfactual trajectory and highlight the need for future research studying cases that were not affected by the covid pandemic and thus allow for more flexible donor pool construction.

Practically, the results indicate that high-profile architectural projects can produce lasting tourism benefits when incorporated into broader urban strategies, although outcomes depend on the specific local context (Alaily-Mattar and Thierstein, 2020). Decision-makers must weigh the high upfront costs against potential macroeconomic returns, noting that success is not guaranteed. Beyond economic impact, megaprojects are frequently discussed critically: they can reflect and reinforce power structures, serving the symbolic and material interests of dominant actors, or legitimize neoliberal agendas where the spectacle of innovation conceals exclusionary logics (Novy and Colomb, 2019; Balke et al., 2018). In contrast, other recent work suggests that such buildings may have positive social effects by serving as ‘field-configuring places’ (Capron and Delacour, 2024) or sites of institutional change (Johannes Dreher and Thiel, 2022). Comparative case studies investigating the mechanisms underlying the success or failure of these projects would make for a valuable contribution to this field. Finally, beyond economic impact, there are also social and cultural aspects to exceptional architecture, which are frequently discussed critically: Megaprojects can reflect and reinforce power structures, serving the symbolic and material interests of dominant actors. They may also be used to legitimize neoliberal agendas and promote ‘post-political’ forms of urban governance, where the spectacle of innovation and strategic planning conceals the exclusionary and commercial logics underlying development (Novy and Colomb, 2019; Balke et al., 2018; E. Häusler and J. Häusler, 2024; Bassols, 2019). In contrast, other recent work suggests that such buildings may have positive social and cultural effects by serving as ‘field-configuring places’ (Capron and Delacour, 2024), or by functioning as sites of innovation and institutional change (Johannes Dreher and Thiel, 2022; Lawrence and Dover, 2015). Research reconciling economic effects and their contribution to and interaction with urban social structure and culture thus seems like another worthwhile avenue for future studies.

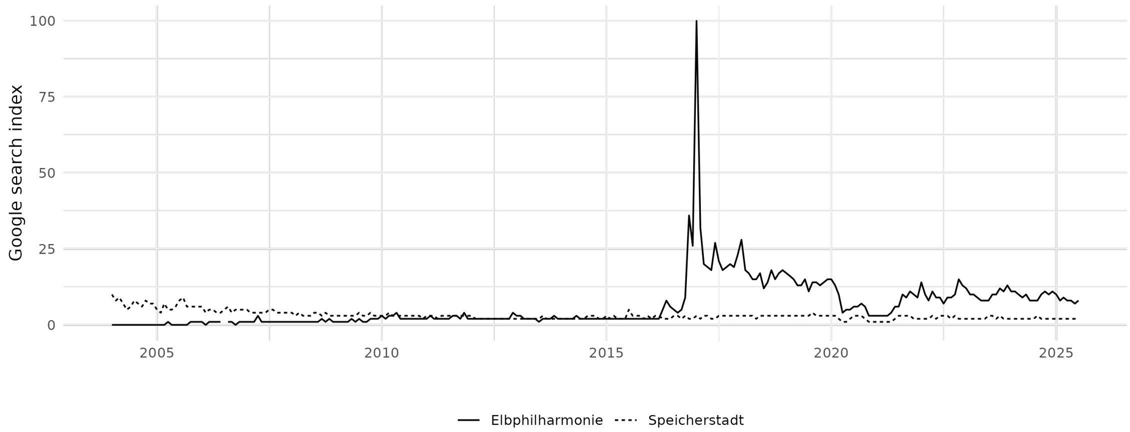

Figure 5: Google trends for search terms ‘Elbphilharmonie’ and ‘Speicherstadt’. Data retrieved from Google Trends (2025).

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